

Phase 1 — All-462-Pair LaBSE Fine-tuning

Training Setup

Parameter	Value
Base model	sentence-transformers/LaBSE
Training pairs	All 462 directed Indic-Indic pairs (22 languages)
Examples per pair	500
Total pair rows	231,000 (207,900 train / 23,100 val, 90/10 split)
Batch size	32
Learning rate	2e-6
Epochs	5

This differs from the earlier single-pair fine-tuning setup (900 examples per pair, 5e-6 learning rate) — Phase 1 uses fewer examples per pair (500) but covers all 462 pairs simultaneously with in-batch negatives drawn across multiple language pairs in each training step, which is the key mechanism intended to prevent the cross-pair interference seen in the earlier combined/mixed models.

In-Domain vs Unseen-Domain Performance

The training validation (IN22-Gen, same domain as training) showed a strong cosine gap of 0.6406 at the best epoch. When evaluated on IN22-Conv a completely unseen conversational domain ,performance drops to a cosine gap of 0.4416.

Evaluation	Cosine Gap
IN22-Gen validation (in-domain, epoch 4)	0.6406
IN22-Conv evaluation (unseen domain)	0.4416
Generalisation gap	0.1991
LaBSE baseline (IN22-Conv)	0.3670

Critically, even after this generalisation gap, the Phase 1 model still beats the LaBSE baseline on unseen data by +0.0746 cosine gap — confirming the improvement is real and not an artifact of overfitting to IN22-Gen.

Phase 1 vs All 14 Previous Models — Full Ranking (IN22-Conv)

Ranked by cosine gap, highest to lowest.

Model	Cosine Gap	Sensitivity	Specificity	Accuracy@1
bottom_single_5_ben_to_mni	0.4624	0.8867	0.9320	0.5796
bottom_single_4_kan_to_sat	0.4573	0.8842	0.9321	0.5788
top_single_5_guj_to_urd	0.4553	0.8336	0.9257	0.5741
top_single_2_urd_to_guj	0.4549	0.8331	0.9265	0.5757
top_single_4_guj_to_mar	0.4534	0.8314	0.9261	0.5748
top_single_3_ben_to_kan	0.4506	0.8313	0.9260	0.5728
labse_all_462_directed_pairs_balanced	0.4416	0.9330	0.9132	0.5983
top_single_1_kan_to_tel	0.4401	0.8260	0.9265	0.5730
bottom_single_3_sat_to_hin	0.4232	0.8675	0.9353	0.5855
bottom_single_2_sat_to_urd	0.4199	0.8668	0.9361	0.5853
bottom_single_1_sat_to_npi	0.4135	0.8636	0.9360	0.5837
top5_all_directed_pairs	0.4007	0.8192	0.9272	0.5763
labse_baseline	0.3670	0.8069	0.9180	0.5654
mixed_top5_bottom5	0.3412	0.8929	0.8592	0.4916
bottom5_all_directed_pairs	0.2286	0.8893	0.7750	0.3945

Key Observations

1. Highest Sensitivity Among All Models

Phase 1 achieves the highest sensitivity (0.9330) among all 15 models evaluated. At the same time, its specificity remains close to the LaBSE baseline (0.9132 vs 0.9180). This shows that the model is much better at identifying true translation pairs while still rejecting most incorrect pairs.

2. Good Performance Across All 22 Languages

Although six specialised single-pair models achieve a higher cosine gap, they are designed for only one or two language pairs. Phase 1 is the only model trained on all 462 directed language

pairs across 22 languages, making it the most balanced and general-purpose model in the comparison.

3. Successful Multi-Pair Training

Previous models trained on multiple language pairs together showed poor performance. Phase 1 avoids this issue and improves over the LaBSE baseline by +0.0746 cosine gap and +0.0329 accuracy@1. This suggests that using in-batch cross-lingual negatives is effective for large-scale multilingual training.

4. Strong Performance on Unseen Data

The model performs better on the training domain than on unseen data, showing a generalisation gap. However, it still clearly outperforms the original LaBSE model on the unseen IN22-Conv dataset, which is the most important measure for real-world use.

